# Integration with SymPy and SciPy

## Exact and numerical integrals in one dimension

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# Two routes to a definite integral

$$\int_a^b f(x)\,dx \;=\; F(b) - F(a), \qquad F'(x) = f(x)$$

## Exact: SymPy

Find an antiderivative $F$.

✓ a formula, no rounding

✗ not every $f$ has one

## Numerical: SciPy

Add up $f$ at sample points.

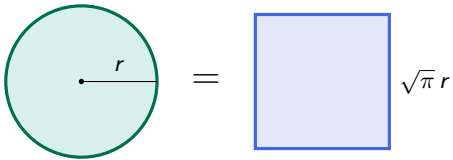
✓ always gives a number

✗ only approximate: check the error

Same idea as derivatives: **formula** vs. **data**.

# Why "quadrature"?

In numerical methods, **integration** is called **quadrature**.

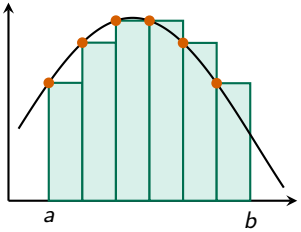



**Quadrature** (ancient geometry): find an area by constructing a **square** of equal area.

Numerical quadrature: $\int_a^b f(x)\,dx \approx \sum_i w_i\, f(x_i)$

# Riemann sums: rectangles

Left Riemann sum

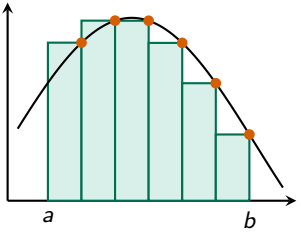



height from the **left** end

$$h\sum_{i=0}^{n-1} f(x_i)$$

$$E \approx \frac{(b-a)\,h}{2}\,f'(\xi)$$

Right Riemann sum

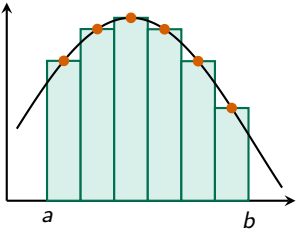



height from the **right** end

$$h\sum_{i=1}^{n} f(x_i)$$

$$E \approx -\frac{(b-a)\,h}{2}\,f'(\xi)$$

Midpoint rule




height from the **middle**

$$h\sum_{i=0}^{n-1} f\left(x_i + \tfrac{h}{2}\right)$$

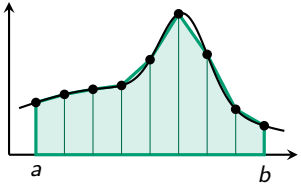
$$E \approx \frac{(b-a)\,h^2}{24}\,f''(\xi)$$

$n$ intervals of width $h = \frac{b-a}{n}$ : left/right $\sim h$, midpoint $\sim h^2$

# Numerical quadrature: three rules

Split $[a, b]$ into pieces, fit a simple polynomial on each, and integrate it exactly.

Trapezoidal, $n = 8$

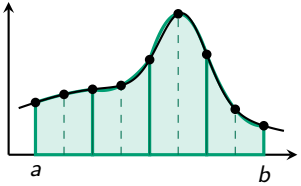



a line on each piece

$$\frac{h}{2}(f_0 + 2f_1 + \cdots + f_n)$$

$$E \sim h^2$$

Simpson, $n = 8$

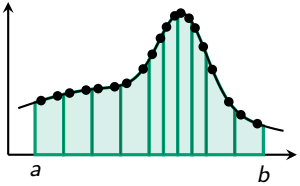



a parabola on each pair

$$\frac{h}{3}(f_0 + 4f_1 + 2f_2 + \cdots + f_n)$$

$$E \sim h^4 \quad (n \text{ even})$$

Gauss (quad)




best points $x_i$ and weights $w_i$

$$\sum_{i=1}^{n} w_i f(x_i)$$

exact for degree $\leq 2n - 1$

Trapezoid, Simpson: $n = \#\text{intervals}$, $h = \frac{b-a}{n}$ Gauss: $n = \#\text{evaluation points}$